MA6.302 (Monsoon 2026)
Probability and Statistics
Assignment 1

Instructor: Naresh Manwani

Deadline: 10:00 AM, 25th August, 2026

Total Marks: 100

Instructions

1. **All questions are compulsory.** Show all relevant working and clearly justify each step. State the axiom, definition, theorem, or result being used wherever applicable.
2. The assignment must be submitted in **both hard copy and soft copy**. The hard copy must be **entirely handwritten**; typed or printed submissions will not be accepted. The soft copy must be uploaded to Moodle as a single PDF named `rollnumber_a1.pdf`.
3. **Group discussion and consultation of reference materials are encouraged** for developing understanding and solving problems. However, the final solution and write-up must be **your own individual work**.
4. If you substantially use any textbook, lecture notes, website, paper, or other reference material while preparing your solution, you must **properly cite/reference the source** in your submission.
5. **Plagiarism or copying will not be tolerated.** Any submission found to contain plagiarised or copied work may receive **zero marks for the assignment/component** and, in serious or repeated cases, the penalty may be extended to an **F grade in the course**, as per course/institute policy.

Question 1. *[10 marks]* **Continuity of probability.** Let $A_1, A_2, A_3, \ldots$ be events.

(a) *[5]* Suppose the sequence is increasing: $A_1 \subset A_2 \subset A_3 \subset \cdots$. Prove

$$\mathrm{P}\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \lim_{n\to\infty} \mathrm{P}(A_n).$$

Hint: write the infinite union as a countable disjoint union, then use the additivity axiom.

(b) *[5]* Suppose the sequence is decreasing: $A_1 \supset A_2 \supset A_3 \supset \cdots$. Prove

$$\mathrm{P}\Big(\bigcap_{i=1}^{\infty} A_i\Big) = \lim_{n\to\infty} \mathrm{P}(A_n).$$

Hint: apply (a) to the complements.

Question 2. *[8 marks]* **Which extra information actually helps.** A coin with $\mathrm{P}(H) = p \in (0, 1]$ is tossed twice. Let

$$A = \{\text{both heads}\}, \qquad B = \{\text{first toss is heads}\}, \qquad C = \{\text{at least one heads}\}.$$

(a) *[4]* Compute $\mathrm{P}(A \mid B)$ and $\mathrm{P}(A \mid C)$ in terms of p.

(b) *[3]* Prove $\mathrm{P}(A \mid B) \geq \mathrm{P}(A \mid C)$, with equality if and only if $p = 1$.

(c) *[1]* Show that $A \subset B \subset C$. Using this chain of inclusions, deduce $\mathrm{P}(A \mid B) \geq \mathrm{P}(A \mid C)$.

Question 3. *[14 marks]* **Three ways of learning that there is a girl.** A family has $n \geq 2$ children. Each child is a boy or a girl with equal probability $1/2$, independently of the others.

(a) *[4]* You ask the father: "Do you have at least one daughter?" He says yes. Find

$$\mathrm{P}(\text{all } n \text{ children are girls} \mid \text{at least one girl}).$$

(b) *[6]* You ask instead: "Do you have at least one daughter named Lilia?" He says yes. Assume that a girl is named Lilia with probability $\alpha \in (0, 1)$, independently of everyone else, and that no boy is named Lilia. Find

$$\mathrm{P}(\text{all } n \text{ children are girls} \mid \text{at least one girl named Lilia})$$

in terms of n and α. Then find the limit of this probability as $\alpha \to 0$, and compare it with (a).

(c) *[4]* A child is selected uniformly at random from the n children, and you observe that this child is a girl. Find

$$\mathrm{P}(\text{all } n \text{ children are girls} \mid \text{the selected child is a girl}).$$

Does this calculated probability match the exact result from part (a) or the limit calculated in part (b)? State which one it matches.

Question 4. *[10 marks]* **A box with a two-headed coin.** A box contains two coins: a fair coin, and a two-headed coin ($\mathrm{P}(H) = 1$). A coin is chosen uniformly at random and tossed independently n times ($n \geq 1$).

(a) *[3]* Find the probability that the first n tosses are all heads.

(b) *[5]* Given that the first n tosses are all heads, find the probability that toss $n+1$ is also a head.

(c) *[2]* What is the limit of the answer in (b) as $n \to \infty$? State whether this limit exactly equals the probability of heads for the fair coin or the two-headed coin.

Question 5. *[10 marks]* **Pairwise is not mutual.** A fair coin is tossed three times. Define

$$\begin{aligned} A &= \{\text{tosses 1 and 2 agree}\}, \\ B &= \{\text{tosses 2 and 3 agree}\}, \\ C &= \{\text{tosses 1 and 3 agree}\}. \end{aligned}$$

(a) *[2]* Find $\mathrm{P}(A)$, $\mathrm{P}(B)$, and $\mathrm{P}(C)$.

(b) *[4]* Show that A, B, C are pairwise independent.

(c) *[4]* Show that they are **not** mutually independent.

Question 6. *[10 marks]* **Independence and conditioning cut both ways.**

(a) *[4]* Let A, B, C be independent events with $\mathrm{P}(C) > 0$. Prove

$$\mathrm{P}(A \cap B \mid C) = \mathrm{P}(A \mid C)\,\mathrm{P}(B \mid C).$$

(b) *[3]* Toss a fair coin twice. Let $H_1 = \{\text{first toss is } H\}$, $H_2 = \{\text{second toss is } H\}$, and $E = \{\text{the two tosses differ}\}$. Show that $H_1 \perp H_2$, but that H_1 and H_2 are **not** conditionally independent given E.

(c) *[3]* Two urns: urn 1 contains 3 red and 1 blue; urn 2 contains 1 red and 3 blue. An urn is chosen uniformly at random, then two balls are drawn **with replacement**. Let R_1 (resp. R_2) be the event that the first (resp. second) ball is red, and let U_1 be the event that urn 1 was chosen. Show that $R_1 \not\perp R_2$, but that $R_1 \perp R_2 \mid U_1$.

Question 7. *[8 marks]* **Odd number of successes.** Let $X \sim \mathrm{Binomial}(n, p)$ with $0 < p < 1$.

(a) *[2]* Write $p_X(k)$ and state why $\sum_{k=0}^{n} p_X(k) = 1$.

(b) *[6]* Prove

$$\mathrm{P}(X \text{ is odd}) = \frac{1 - (1 - 2p)^n}{2}.$$

Hint: expand $(q + p)^n$ and $(q - p)^n$ where $q = 1 - p$.

Question 8. *[10 marks]* **How many packets until the message decodes.** Packets are sent one by one. Each packet arrives correctly with probability $p \in (0, 1)$, independently of the others (otherwise it is lost). The receiver can decode as soon as it has k correct packets, where $k \geq 1$ is fixed. Let X be the number of packets the sender must transmit until decoding succeeds. (So $X \geq k$.)

Do not quote a named distribution other than Bernoulli, binomial, or geometric. Derive (b) from (a).

(a) *[3]* Let E_1 be the event "exactly $k - 1$ of the first $n - 1$ packets are correct" and E_2 be the event "packet n is correct". Show mathematically that $\{X = n\} = E_1 \cap E_2$. (Assume $n \geq k$.)

(b) *[5]* Find $p_X(n)$ for $n = k, k + 1, k + 2, \ldots$.

(c) *[2]* Check that your formula reduces to the geometric PMF from Lecture 4 when $k = 1$.

Question 9. *[8 marks]* **Median.** A number m is called a **median** of a discrete random variable X if

$$\mathrm{P}(X \geq m) \geq \tfrac{1}{2} \qquad \text{and} \qquad \mathrm{P}(X \leq m) \geq \tfrac{1}{2}.$$

A median need not be unique. Find *all* medians in each case.

(a) *[2]*

$$p_X(k) = \begin{cases} 0.4 & k = 1, \\ 0.3 & k = 2, \\ 0.3 & k = 3, \\ 0 & \text{otherwise.} \end{cases}$$

(b) *[2]* X is the result of rolling a fair six-sided die.

(c) *[4]* $X \sim \text{Geometric}(p)$, $0 < p < 1$ (number of trials until the first success). Express the median(s) in terms of p.
Hint: $\mathrm{P}(X \leq m) = 1 - (1-p)^m$ and $\mathrm{P}(X \geq m) = (1-p)^{m-1}$.

Question 10. *[12 marks]* **A casino game.** A fair coin is tossed until the first tail. If the first tail occurs on toss k, the player is paid 2^{k-1} dollars. Let X be the payment, and let K be the toss on which the first tail appears, so $K \sim \text{Geometric}(1/2)$ and $X = 2^{K-1}$.

(a) *[3]* Find $\mathrm{P}(X > 65)$. (This part does not use averages.)

For a discrete random variable Z taking values in a countable set, define

$$\mathcal{A}(Z) := \sum_z z\, p_Z(z),$$

allowing the value $+\infty$ if the series diverges. This is only a weighted sum of the PMF. Do not use any other property.

(b) *[5]* Compute $\mathcal{A}(X)$. If a "fair entry fee" is strictly defined as a finite amount equal to $\mathcal{A}(X)$, does a fair entry fee exist for this game? State Yes or No.

(c) *[4]* The casino will pay at most 2^{30} dollars: let $Y = \min(X, 2^{30})$. Compute $\mathcal{A}(Y)$.